

Name: _____

Teacher: BMM RABS RPN



CRANBROOK
SCHOOL

+ SOLUTIONS

Year 11 Mathematics

Preliminary Examination

September 2nd, 2008

Instructions

- Attempt Questions 1-7
- All questions are of equal value
- Answer each question in a new booklet
- All necessary working should be shown in every question
- Board approved calculators are allowed in all sections

Time Allowed: 2 hours

Total Marks: 84

Question 1 (12 Marks)

Start a new booklet

Marked by BMM

- (a) In a weekend garage sale, $\frac{3}{5}$ of the items were sold on Saturday. $\frac{2}{3}$ of the remaining items were then sold on Sunday. What percentage of the items remained unsold at the end of the weekend? 2
- (b) Simplify by removing the parentheses:
- (i) $\sqrt{8}(\sqrt{2} - \sqrt{3})$ 1
- (ii) $(2\sqrt{5} - \sqrt{3})^2$ 1
- (c) Solve: $x^2 + 9x = 10$ 2
- (d) Factorise completely:
- (i) $m^3 + 10m^2 + 25m$ 1
- (ii) $x^3 - \frac{1}{8}$ 1
- (e) Simplify: $\frac{1}{x+5} - \frac{1}{x-5}$ 2
- (f) Evaluate $255(0.0024)^{10}$ giving your answer in scientific notation correct to 3 significant figures. 2

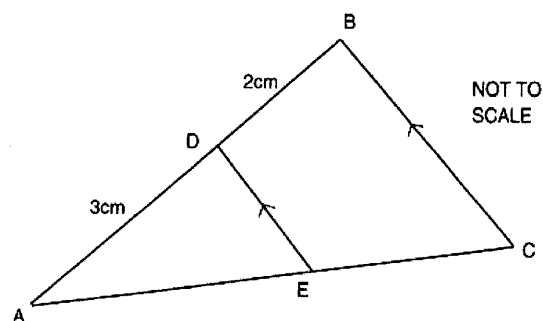
End of Question 1

Question 2 (12 Marks)

Start a new booklet

Marked by BMM

- (a) Solve: $|3 - 2x| > 5$ 2
- (b) Solve: $\sin \beta - \frac{\sqrt{3}}{2} = 0$ for $0^\circ \leq \beta \leq 360^\circ$ 2
- (c) If $\frac{2}{3 - \sqrt{7}} = a + b\sqrt{7}$, find the values of a and b . 2
- (d) Copy or trace the diagram into your booklet. 4



In $\triangle ABC$, points D and E lie on lines AB and AC respectively, such that DE is parallel to BC . If $AD = 3$ cm and $BD = 2$ cm, find the ratio $DE : BC$, giving reasons.

- (e) Solve the following pair of simultaneous equations: 2
- $$\begin{aligned} 2x - y &= -8 \\ 3x + 2y &= -5 \end{aligned}$$

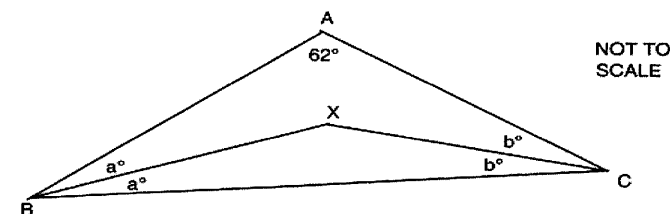
End of Question 2

Question 3 (12 Marks)

Start a new booklet

Marked by RABS

- (a) Differentiate each of the following:
- (i) $\frac{3x}{x^2 - 2x}$ 2
- (ii) $(2 - 3x)^7$ 1
- (iii) $x\sqrt{1 - x}$ 2
- (iv) $\sqrt[3]{x} - \frac{1}{x}$ 2
- (b) Shade the region on the number plane where the following inequalities hold simultaneously: 2
- $$\begin{aligned} y &\geq x^2 - 1 \\ y &\leq x + 1 \end{aligned}$$
- (c) Copy or trace the diagram into your booklet.



In the diagram above, XB and XC bisect $\angle ABC$ and $\angle ACB$ respectively. $\angle BAC = 62^\circ$. Find the size of $\angle BXC$.

3

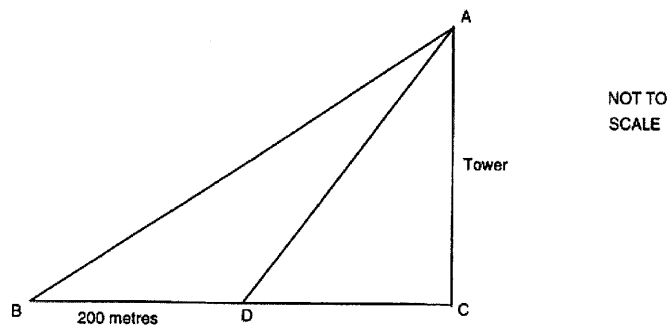
End of Question 3

- (a) Solve, leaving your answers in exact form: 3

$$x^6 + 6x^3 - 16 = 0$$

- (b) Find the equation of the normal to the curve $y = x^3 - 3x + 2$ at the point (2, 4). 3

(c)



The diagram above shows a vertical tower AC. Points B, C and D are in a straight line on level ground. The distance from B to D is 200m.

A surveyor found that the angle of elevation to the top of the tower from point B was 38° . She then moved to point D and measured the angle of elevation as 54° .

- (i) Copy or trace the diagram and fill in all the information given. 1

- (ii) Show that the length $AD = \frac{200 \sin 38^\circ}{\sin 16^\circ}$ 2

- (iii) Calculate the height of the tower. 1

- (d) A circle has the equation $x^2 - 2x + y^2 + 6y = 6$. By writing the equation in the form $(x - a)^2 + (y - b)^2 = c$, or otherwise, find the radius of the circle and the co-ordinates of the centre. 2

Question 5 (12 Marks)

Start a new booklet

Marked by RPN

- (a) Given that α and β are the roots of the quadratic equation $3x^2 - 5x - 1 = 0$, find the values of:
- (i) $\alpha + \beta$ 1
 - (ii) $\alpha\beta$ 1
 - (iii) $\alpha^2 + \beta^2$ 2
- (b) A ship sails from point A on a bearing of $237^\circ T$ for a distance of 423 km. The ship then turns and sails due South to point C. The bearing of A from C is found to be $41^\circ T$.
- (i) Draw a diagram showing the above information. 1
 - (ii) Find the size of $\angle BAC$. 1
 - (iii) Calculate the total distance sailed by the ship. 2
- (c) A parabola has the equation $4y = x^2 + 4x + 8$.
- (i) Find its focal length. 1
 - (ii) Find the coordinates of its vertex. 1
 - (iii) State the equation of the axis of symmetry. 1
 - (iv) Draw a neat sketch of the parabola. 1

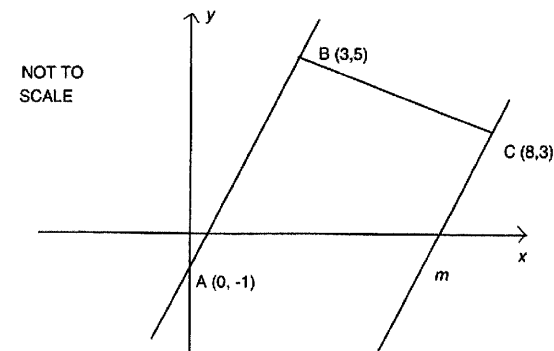
End of Question 5

Question 6 (12 Marks)

Start a new booklet

Marked by RPN

(a)

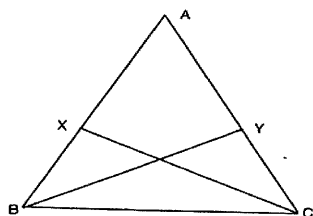


In the diagram, $A(0, -1)$, $B(3, 5)$ and $C(8, 3)$ are points on the number plane. Copy or trace the diagram into your booklet.

- (i) Find the length of AB . 1
- (ii) Find the equation of AB . 2
- (iii) Find the equation of the line m passing through C parallel to AB . 2
- (iv) The point D lies on m such that BC is parallel to AD . Find the coordinates of D . 2
- (v) What type of quadrilateral is $ABCD$? Give reasons. 2
- (vi) Find the perpendicular distance from C to AB . 2
- (vii) Find the area of $ABCD$. 1

End of Question 6

(a)

NOT TO
SCALE

In the diagram above, $\triangle ABC$ is isosceles with $AB = AC$. Points X and Y lie on AB and AC respectively such that $AX = AY$.
Copy or trace the diagram into your booklet.

(i) Prove that $\triangle BXC \equiv \triangle CYB$ 4

(ii) Hence or otherwise prove that $\angle XBY = \angle YCX$ 1

(b) A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} \frac{1}{x} & \text{for } x > 0 \\ 1 & \text{for } x = 0 \\ 2^x & \text{for } x < 0 \end{cases}$$

(i) Draw a neat sketch of the graph $y = f(x)$. 3

(ii) Evaluate: $f(2) + f(0) + f(-2)$ 2

(c) Simplify: $2\cos^2 B + 3\sin^2 B - 2$ 2

End of Question 7
End of Examination

QUESTION 1

$$(a) \quad \frac{3}{5} + \frac{2}{3} \times \frac{2}{5} = \frac{13}{15} \quad \checkmark$$

$$\text{UNSOLD} = \frac{2}{15} = \underline{13.\dot{3}\%} \quad \checkmark$$

$$(b) \quad (i) \quad \sqrt{16} - \sqrt{24} = \underline{4 - 2\sqrt{6}} \quad \checkmark$$

$$(ii) \quad (2\sqrt{5} - \sqrt{3})(2\sqrt{5} - \sqrt{3})$$

$$= 4 \times 5 - 4\sqrt{15} + 3$$

$$= \underline{23 - 4\sqrt{15}} \quad \checkmark$$

$$(c) \quad x^2 + 9x - 10 = 0$$

$$(x+10)(x-1) = 0 \quad \checkmark$$

$$\therefore \underline{x = -10, 1} \quad \checkmark$$

$$(d) \quad (i) \quad m(m^2 + 10m + 25)$$

$$= \underline{m(m+5)^2} \quad \checkmark$$

$$(ii) \quad x^3 - \frac{1}{8} = \underline{(x - \frac{1}{2})(x^2 + \frac{1}{2}x + \frac{1}{4})} \quad \checkmark$$

$$(e) \quad \frac{(x-5)}{(x+5)(x-5)} - \frac{(x+5)}{(x+5)(x-5)} \quad \checkmark$$

$$= \underline{\frac{-10}{x^2-25}} \quad \checkmark$$

$$(f) \quad = \underline{1.62 \times 10^{-24}} \quad \checkmark$$

QUESTION 2

$$(a) \quad 3 - 2x > 5$$

$$-2x > 2$$

$$\underline{x < -1} \quad \checkmark$$

$$-(3 - 2x) > 5$$

$$-3 + 2x > 5$$

$$2x > 8$$

$$\underline{x > 4} \quad \checkmark$$

$$(b) \quad \sin \beta = \frac{\sqrt{3}}{2}$$

$$\beta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$= 60^\circ$$

$\sin$ positive in 1st & 2nd Quadrants.

$$\therefore \underline{\beta = 60^\circ \neq 120^\circ} \rightarrow (180^\circ - \theta)$$

$$\textcircled{c} \quad \frac{2}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}} = \frac{6+2\sqrt{7}}{9-7}$$

$$= \frac{6+2\sqrt{7}}{2}$$

$$= 3+\sqrt{7}$$

$$\therefore \underline{a=3, b=1}$$

$\textcircled{d}$ IN $\triangle ADE \neq \triangle ABC \dots$

- $\angle ADE = \angle ABC$ (corresponding $\angle$ s on $\parallel$ lines) ✓
- $\angle AED = \angle ACB$ (corresponding $\angle$ s on $\parallel$ lines) ✓
- $\angle DAE = \angle BAC$ (common)

$$\therefore \triangle ADE \parallel \triangle ABC \text{ (AAA)} \checkmark$$

$$\therefore DE:BC = AD:AB$$

$$= \underline{3:5} \checkmark$$

$$\textcircled{e} \quad 4x - 2y = -16 \text{ --- } \textcircled{3}$$

$$3x + 2y = -5 \text{ --- } \textcircled{2}$$

$$\textcircled{3} + \textcircled{2} \quad 7x = -21$$

$$\underline{x = -3} \checkmark$$

Sub $x = -3$ into $\textcircled{3}$

$$-12 - 2y = -16$$

$$-2y = -4$$

$$\underline{y = 2} \checkmark$$

QUESTION 3

$$\textcircled{a} \quad \textcircled{i} \quad u = 3x \quad v = x^2 - 2x$$

$$u' = 3 \quad v' = 2x - 2$$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{3(x^2 - 2x) - 3x(2x - 2)}{(x^2 - 2x)^2} \checkmark$$

$$= \frac{3x^2 - 6x - 6x^2 + 6x}{(x^2 - 2x)^2}$$

$$= \frac{-3x^2}{(x^2 - 2x)^2} \checkmark$$

ii) $y' = 7(2 - 3x)^6 \cdot (-3)$

$$= -21(2 - 3x)^6 \checkmark$$

iii) $u = x \quad v = \sqrt{1-x} = (1-x)^{1/2}$

$$u' = 1 \quad v' = \frac{1}{2}(1-x)^{-1/2} \cdot -1$$

$$= -\frac{1}{2}(1-x)^{-1/2}$$

$$y' = vu' + uv'$$

$$= (1-x)^{1/2} + -\frac{x}{2}(1-x)^{-1/2} \checkmark$$

$$= \sqrt{1-x} - \frac{x}{2\sqrt{1-x}}$$

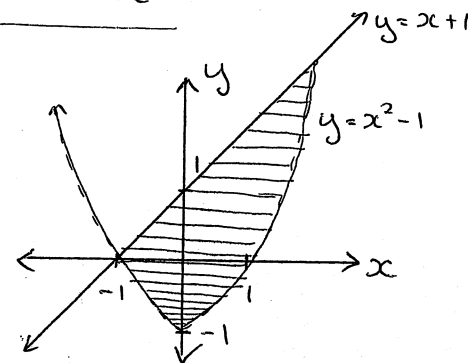
$$= \frac{2(1-x) - x}{2\sqrt{1-x}} = \frac{2-3x}{2\sqrt{1-x}} \checkmark$$

iv) $y = x^{1/3} - x^{-1}$

$$y' = \frac{1}{3}x^{-2/3} + x^{-2} \checkmark$$

$$= \frac{1}{3\sqrt[3]{x^2}} + \frac{1}{x^2} \checkmark$$

b)



• correct graphs $\checkmark$
• regions $\checkmark$

c) $62^\circ + 2a + 2b = 180^\circ \checkmark$

$$2a + 2b = 118^\circ$$

$$a + b = 59^\circ \checkmark$$

$$\angle BXC + a + b = 180^\circ$$

$$\angle BXC + 59^\circ = 180^\circ$$

$$\therefore \angle BXC = 121^\circ \checkmark$$

QUESTION 4

① Let $X = x^3$

$$\therefore X^2 + 6X - 16 = 0 \checkmark$$

$$(X+8)(X-2) = 0$$

$$\therefore \underline{X = -8 \text{ or } 2} \checkmark$$

$$\therefore x^3 = -8 \text{ or } 2$$

$$\underline{x = -2 \text{ or } \sqrt[3]{2}} \checkmark$$

⑥ $y' = 3x^2 - 3$

$$\text{Gradient} = 3 \times 2^2 - 3 \quad (\text{sub in } x=2)$$

$$= 9 \checkmark$$

$$\therefore \text{Normal} = -\frac{1}{m_1} = -\frac{1}{9} \checkmark$$

pt/grad. formula:

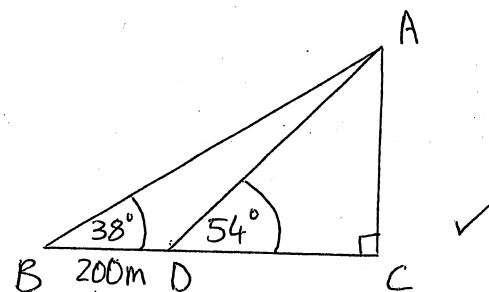
$$y - 4 = -\frac{1}{9}(x - 2)$$

$$9y - 36 = -x + 2$$

$$\underline{x + 9y - 38 = 0} \checkmark$$

②

①



②

$$54^\circ = 38^\circ + \angle BAD \quad (\text{ext. angle theorem})$$

$$\therefore \underline{\angle BAD = 16^\circ} \checkmark$$

Sine rule:

$$\frac{AD}{\sin 38^\circ} = \frac{200}{\sin 16^\circ}$$

$$AD = \frac{200 \sin 38^\circ}{\sin 16^\circ} \checkmark$$

③

In $\triangle ACD$,

$$\sin 54^\circ = \frac{AC}{AD}$$

$$= \frac{AC}{AD} \Rightarrow AC = AD \sin 54^\circ$$

$$AC = \frac{200 \sin 38^\circ \cdot \sin 54^\circ}{\sin 16^\circ}$$

$$= \underline{361.4028 \dots \text{ m}} \checkmark$$

② COMPLETE THE SQUARE:

$$x^2 - 2x + 1 + y^2 + 6y + 9 = 6 + 1 + 9$$

$$(x-1)^2 + (y+3)^2 = 16$$

$$\therefore \text{radius} = \sqrt{16} = 4 \checkmark$$

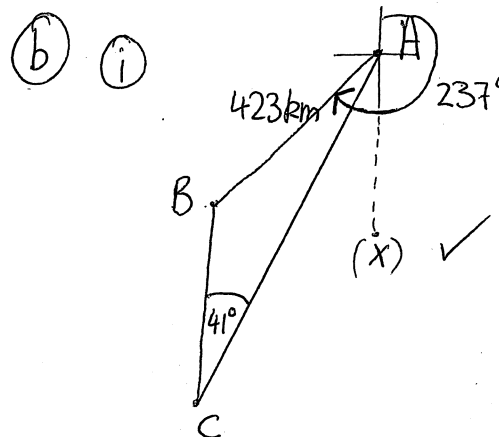
$$\text{centre} = \underline{(1, -3)} \checkmark$$

QUESTION 5

② ① $\alpha + \beta = \frac{-b}{a} = \underline{\frac{5}{3}} \checkmark$

② ② $\alpha\beta = \frac{c}{a} = \underline{\frac{-1}{3}} \checkmark$

② ③ $(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$
 $\therefore \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \checkmark$
 $= \left(\frac{5}{3}\right)^2 - 2\left(-\frac{1}{3}\right)$
 $= \underline{3\frac{4}{9}} \checkmark$



② ② Add point X above to help.

$\angle BCA = \angle CAX$ (alternate $\angle$ s on // lines)

$$\therefore \angle CAX = 41^\circ$$

$$\angle BAX = 237^\circ - 180^\circ = 57^\circ$$

$$\therefore \angle BAC = 57^\circ - \angle CAX^\circ = \underline{16^\circ} \checkmark$$

② ③ Sine rule:

$$\frac{BC}{\sin 16^\circ} = \frac{423}{\sin 41^\circ} \Rightarrow BC = \frac{423 \sin 16^\circ}{\sin 41^\circ} = \underline{177.7 \text{ km}} \checkmark$$

$$\therefore \underline{\text{TOTAL DISTANCE}} = \underline{600.7 \text{ km}} \checkmark$$

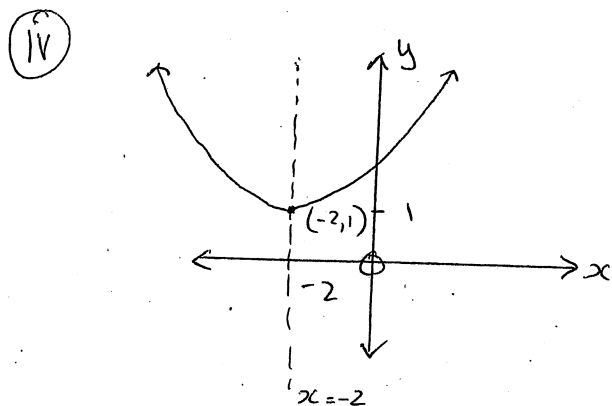
② (i) $4y - 4 = x^2 + 4x + 4$

$4(y-1) = (x+2)^2$

$\therefore a = \frac{4}{4} = 1 \checkmark$

(ii) FROM THE EQ'N:
 $(-2, 1)$ $\checkmark$

(iii) AS x-COORD IS 2, THE
AXIS OF SYM. IS $x = -2$ $\checkmark$



QUESTION 6

(i) $AB = \sqrt{(3-0)^2 + (5+1)^2}$
 $= \sqrt{9 + 36}$
 $= \sqrt{45} \text{ UNITS OR } 3\sqrt{5} \checkmark$

(ii) $AB \text{ gradient} = \frac{5+1}{3-0} = \frac{6}{3} = 2 \checkmark$

USING PT./GRAD:

$y - 5 = 2(x - 3)$

$y - 5 = 2x - 6$

$2x - y - 1 = 0$ $\checkmark$

(iii) $AB \text{ gradient} = m \text{ gradient (11 lines)}$

USING PT./GRAD:

$y - 3 = 2(x - 8) \checkmark$

$y - 3 = 2x - 16$

$2x - y - 13 = 0$ $\checkmark$

$$\textcircled{\text{iv}} \text{ BC gradient} = \frac{3-5}{8-3} = -2/5$$

USING PT/GRAD ON AD:

$$y+1 = -2/5(x-0)$$

$$5y+5 = -2x$$

$$\underline{2x + 5y + 5 = 0} \checkmark$$

SOLVE SIM. WITH EQ'N OF M:

$$2x + 5y + 5 = 0 \text{ --- } \textcircled{1}$$

$$2x - y - 13 = 0 \text{ --- } \textcircled{2}$$

$$\textcircled{1} - \textcircled{2}$$

$$6y + 18 = 0 \Rightarrow \text{CO-ORDS} (5, -3) \checkmark$$

$$6y = -18$$

$$y = -3$$

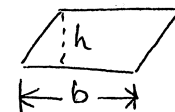
SUB INTO $\textcircled{1} \therefore x = 5$

$\textcircled{\text{v}}$ PARALLELOGRAM. (2 PAIRS OF OPP. SIDES) $\checkmark$

$$\textcircled{\text{vi}} d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|2 \times 8 + (-3) - 1|}{\sqrt{2^2 + 1^2}} = \frac{12}{\sqrt{5}} \text{ UNITS } \checkmark$$

USING
 $2x - y - 1 \neq (8, 3)$

$$\textcircled{\text{vii}} \text{ AREA OF //GRAM} = b \times h$$



$$= 3\sqrt{5} \times \frac{12}{\sqrt{5}}$$

FROM $\textcircled{\text{i}}$ FROM $\textcircled{\text{vi}}$

$$= \underline{36 \text{ UNITS}^2} \checkmark$$

QUESTION 7

$$\textcircled{\text{a}} \textcircled{\text{i}} \cdot \angle XBC = \angle YCB \text{ (base } \angle \text{s of isos.)}$$

Δ are =. $\checkmark$

$$\cdot AB - AX = AC - AY$$

$$\therefore BX = YC \text{ (given)} \checkmark$$

$\cdot BC$ is common

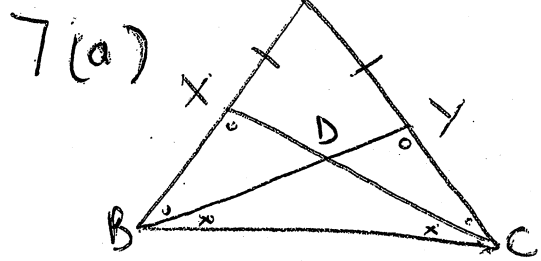
$$\therefore \Delta BXC \equiv \Delta CYB \text{ (SAS)} \checkmark$$

$$\textcircled{\text{ii}} \cdot \angle XCB = \angle YBC \text{ (corresp. } \angle \text{s in congr. } \Delta \text{s)} \checkmark$$

$$\cdot \angle XBC = \angle YCB \text{ (from above)}$$

$$\therefore \angle XBC - \angle YBC = \angle YCB - \angle XCB$$

$$\therefore \underline{\angle XBY = \angle YCX} \checkmark$$



(i) In ΔBXC and CYB

$$\angle XBC = \angle YCB \quad (\text{base } \angle \text{ s of isos } \Delta \text{ are equal})$$

$$BX = YC \quad (\text{as } AB = AC \text{ (given)} \\ AX = AY \text{ (given)} \\ BX = AB - AX \\ YC = AC - AY)$$

BC is common

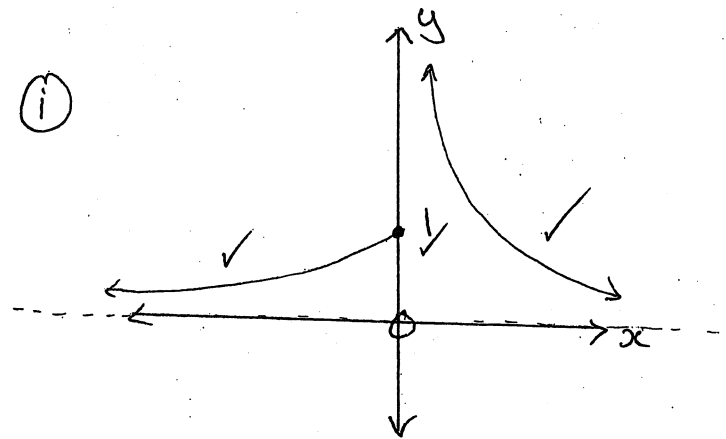
$$\therefore \Delta XBC \equiv \Delta CYB \text{ (SAS)}$$

(ii) $\angle XCB = \angle YBC$ (corresponding sides of congruent Δ s from (i) are equal)

$$\text{But } \angle XBC = \angle YCB \quad (\text{base } \angle \text{ s of isos } \Delta \text{ are equal})$$

$$\therefore \angle XBC - \angle YBC = \angle YCB - \angle XCB \\ \Rightarrow \angle XBY = \angle YCX$$

7(b) (i)



(ii)

$$\int (2) + \int (0) + \int (-2) \\ = \frac{1}{2} + 1 + 2^{-2} \\ = 1\frac{3}{4}$$

(c)

$$2 \cos^2 B + 3 \sin^2 B - 2 \\ = 2 \cos^2 B + 2 \sin^2 B + \sin^2 B - 2 \\ = 2 (\cos^2 B + \sin^2 B) + \sin^2 B - 2 \\ = 2 + \sin^2 B - 2 \\ = \sin^2 B$$

or (ii) In ΔXBD and YCD :

$$\angle CYD = \angle BXD \quad (\text{corr. } \angle \text{ s of congruent } \Delta \text{ s in (i) are equal}) \\ \angle CDY = \angle BDX \quad (\text{vert. opp. } \angle \text{ s are equal})$$

$$\therefore \angle YCD = \angle XBD \quad (\text{sum of } \Delta = 180^\circ) \\ \Rightarrow \angle YCX = \angle XBY$$

7(a) (In Δ s XBD and YCD):

or (ii) $XB = YC$ (corr. sides of congruent Δ s in (i) are equal)

$$\angle XDB = \angle YDC \quad (\text{vert. opp. } \angle \text{ s are equal})$$

$$\angle CXB = \angle BYC \quad (\text{corr. } \angle \text{ s of congruent } \Delta \text{ s in (i) are equal})$$

$$\therefore \Delta XBD \equiv \Delta YCD \text{ (AAS)}$$

$$\therefore \angle XBY = \angle YCX \quad (\text{corr. angles of congruent } \Delta \text{ s are equal})$$

(In Δ s ABY and ACX):

or (ii) $AX = AY$ (given)

$$\angle A \text{ is common} \\ AB = AC \text{ (given)}$$

$$\therefore \Delta ABY \equiv \Delta ACX \text{ (SAS)}$$

$$\Rightarrow \angle XBY = \angle YCX \quad (\text{corr. angles of congruent } \Delta \text{ s are equal})$$